4.3.2. Problem 2:

The IEEE 30 bus 6 generator test system described has been considered in this problem. Operating Cost of each generator is given by:

$$F_i(P_{gi}) = a_i P_{gi}^2 + b_i P_{gi} + d$$

where a, b and d are the fuel-cost-coefficients. p_i is the real power output of the i[th] generator in MW.

The stipulated conditions for the aforementioned cost function are as follows:

a) The aggregate power (P_L) output from the five generators is required to be 2.86 megawatts.

$$\Sigma P_i = P_L = 2.86\ MW$$

b) The generated power must fall within the specified range for each generator.

$$Pgmin < P_i < Pgmax$$

where $\bar{P}$ is max. power generated and $\underline{P}$ min. power generated.

Generator	a	b	d	Pgmin	Pgmax
1	100	200	10	0.05	1.5
2	120	150	10	0.05	1.5
3	40	180	20	0.05	1.5
4	60	100	10	0.05	1.5
5	40	180	20	0.05	1.5
6	100	150	10	0.03	1.5
Operating Cost $/h					